MIL-S-36377B
29 October 1985
SUPERSEDING
MIL-S-36377A
8 February 1977

MILITARY SPECIFICATION

SPREADER, GUTTA-PERCHA, DENTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers spreaders suitable for use in manipulating gutta-percha into dental pulp chambers and root canals.

1.2 Classification. Spreaders shall be of the following styles, as specified (see 6.1):

Style 1 - No. D11
Style 2 - No. D11T
Style 3 - No. MA57

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specifications.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Part and Equipment, Procedures for Packaging of.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24424 - Spreader, Gutta-Percha, Dental, No. D11 and D11T.
24437 - Spreader, Gutta-Percha, Dental, #A57

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 135 - Specification for Seamless Brass Tube.
ASTM E 18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E 92 - Standard Method of Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. The spreaders shall be in accordance with the applicable requirements of GG-I-526, with the options and with the additional requirements specified herein.

3.2 Material.

3.2.1 Spreader point. The spreader operative end shall be fabricated of class 5, type 17-7 PH precipitation hardening stainless steel or class 3, type 302, austenitic stainless steel.

3.2.2 Handles. Handles shall be fabricated of the following materials:

Solid handles. Class 3, type 303 austentic stainless steel, except, as an alternate, the steel may contain a maximum of 3.50 percent of manganese and 0.25 percent of copper; or Class 4, type 416, martensitic stainless steel.

Hollow handles. Chromium over nickel plated brass. Brass shall conform to copper alloy No. 260 or 330 as specified in ASTM B135 for composition only.

3.3 Hardness. After proper stress relieving and passivation, the Rockwell hardness of the spring tempered operative end shall be C40 to C50. Hardness shall be determined as specified in 4.4.1.

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3.4 Style, design and dimensions. The spreader shall be of two-piece construction consisting of a spreader operative end, and a handle. The spreader operative end shall be firmly and permanently secured in the handle. There shall be no movement, loosening, or separation between the operative end and handle when the operative end is subjected to an 85 pound (minimum) pull out force when tested as specified in 4.4.5. The handle shall be straight, solid, and of one piece construction. The finger grip portion of the handle shall be finely serrated or fine diamond knurled. The spreader shall be suitable for manipulating gutta-percha into pulp chambers and root canals. Style 1 (No. D11) and Style 2 (No. D11T) shall conform to Defense Personnel Support Center (DPSC) Drawing 24424 and Style 3 (No. MA57) shall conform to DPSC Drawing 24437, for style, design, and dimensions.

3.4.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.5 Finish. The spreader operative end shall have a smooth, bright or mirror finish and shall be free of defects such as coarse grind marks, cracks, nicks, gouges and burrs, when viewed as specified in 4.3.1. The handle shall have a smooth, mirror or bright polished finish, except for the serrated or knurled finger grip portion. Serrated or knurled finger grip portions shall be free of barbs and shall provide adequate finger grip during use. The proximal end of the handle shall be smoothly rounded. The interface of the shank and handle shall be smoothly blended with no crevices evident.

3.6 Corrosion resistance. The copper sulfate test shall apply to martensitic stainless steel components. The boil test shall apply to the entire spreader (operative ends and all handles).

3.7 Ductility. The operative end of the spreader shall withstand a minimum of two 90 degree cold bends with no signs of rupture or fracture, when tested as specified in 4.4.4.

3.8 Identification marking. In addition to the marking requirements of GG-I-526, each spreader shall be marked with the applicable instrument number. All markings shall be legible and permanent. The word "stainless" shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered trademark identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I, II, and III (see 4.2.1, 4.2.2, and 4.3) and 4.3.1.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table I. Unit of product for examination purposes shall be the unit of issue.

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TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II.

TABLE II. Test criteria.

Characteristic	Reqmt.	Test procedure	Inspection level	AQL
Chemical composition	3.2 and GG-I-526	GG-I-526	S-1 1/	2/
Hardness	3.3 and GG-I-526	4.4.1	S-2	1.5
Pull test	3.4	4.4.5	S-2	2/
Copper sulfate test	3.6	4.4.2 and GG-I-526	S-2	1.0
Boil test	3.6	4.4.3 and GG-I-526	S-2	1.0
Ductility	3.7	4.4.4	S-2	1.5

1/ A sufficient quantity of each lot of metal used in the construction of the spreader may be selected for composition analysis in lieu of sample instrument, provided lot identity has been maintained and can be demonstrated to the Government.

2/ Acceptance number shall be zero for all sample sizes.

4.2.3 Sample unit usage. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

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4.3 Examination. The spreader shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Any spreader in the sample containing one or more defects shall be rejected, and if the number of defective spreaders in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Examination shall include, but shall not be limited to, the defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major A:	
101	Style and design not as specified.
102	Finish on spreader operative end not as specified.
103	Spreader operative end not free of coarse grind marks, cracks, nicks, gouges, burrs or other imperfections.
104	Spreader operative end bent or distorted.
105	Spreader operative end taper not smooth.
Major B:	
151	Spreader operative end dimensions not as specified.
Minor A:	
201	Handle not straight.
202	Finish on handle not as specified.
203	Instrument identification not present, not legible, not permanent.
204	Instrument not permanently and legibly marked with name or trade mark of manufacturer.
205	Instrument not free of dirt, oil, polishing compound, or foreign matter.
206	Interface of the shank and the handle not smoothly blended; shows crevices.
Minor B:	
251	Dimensions (other than 151) not as specified.

4.3.1 Examination under magnification. The instrument shall be examined under four power (4X) magnification for defect 103. Examination under magnification shall include, but shall not be limited to, the defect listed herein.

4.4 Tests. Tests shall be conducted as necessary to determine compliance with specification requirements.

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4.4.1 Hardness test. Hardness shall be determined in accordance with ASTM Standard E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid Hardness).

4.4.2 Copper sulfate test. The copper sulfate test shall be conducted in accordance with GG-I-526.

4.4.3 Boil test. The boil test shall be conducted in accordance with GG-I-526.

4.4.4 Ductility test. The first 1/4 inch of the operative end (operative tip) shall be clamped in a vise or other suitable device, the jaws of which shall be rounded to a radius to 1/16 inch. The spreader operative end shall be bent back and forth over this radius at a uniform rate. Each 90 degrees of travel in one direction shall be considered one bend. Test shall be conducted at room temperature. Examination for rupture or fracture shall be under a minimum of 30 power (30X) magnification.

4.4.5 Pull test. The handle and shank of the operative end shall be securely held in separate chucks or other holding devices. A slowly increasing pull out force, increasing from zero to 85 pounds (minimum), shall be applied to the shank of the operative end. The force shall be maintained at the high value for a minimum of 1 second. The instrument shall then be examined for movement, looseness, or separation of the operative end with respect to the handle. The test shall be repeated for each operative end of the double-ended instrument.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Spreaders shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Spreaders shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit package. Each spreader shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.1.5 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

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5.1.2 Level C. The level C preservation for spreaders shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for spreaders shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Spreaders packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Spreaders packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate pack, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

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5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style of spreader required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Approval of trademark if applicable (see 3.7).

6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

Stock Number	Item Identification
6520-01-061-6452	SPREADER, GUTTA-PERCHA, DENTAL, No. D11T.
6520-01-064-6961	SPREADER, GUTTA-PERCHA, DENTAL, No. D11.
6520-01-126-4033	SPREADER, GUTTA-PERCHA, DENTAL, No. MA57.

Military Interests:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DoD-MB

Agent:

DLA-DH

Review activities:

Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS

Project No. 6520-1826

Location: S2-9-9

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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